

Describing Admissible Strings II

Regular expressions, linear grammars, and regular relations

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Different descriptions expose different generalizations

Question

When do union, concatenation, and repetition suffice—and when must a dependency be carried?

- Regular expressions characterize local repetition and choice.
- Linear grammars coordinate two regions through one continuing nonterminal.
- Regular relations pair input and output strings for bounded-context mappings.

MOLR §§4.3–4.4

Where we are

1. Regular expressions
2. Linear grammars
3. Regular relations

Regular expressions are recursively defined descriptions

For alphabet Σ :

- 1 $\emptyset$, ϵ , and each $a \in \Sigma$ are regular expressions.
- 2 If u, v are regular expressions, so are $u \cup v$, uv , and u^* .
- 3 Nothing else is a regular expression.

Take-away

A recursive definition closes a base vocabulary under specified constructors.

MOLR §4.3

Three operations build every regular language

Union	$a \cup b$	choose a or b
Concatenation	ab	place a before b
Kleene star	a^*	repeat a zero or more times

Guided analysis

Describe binary strings ending in 01. Then describe those containing at least one 1.

Equivalent expressions can differ greatly in intelligibility

$$(u \cup v)^* = (u^* \cup v)^* = u^*(vu^*)^* = (u^*v)^*u^*$$

- Formal equivalence preserves the stringset.
- Style determines which invariant is easiest to see.
- Simplicity is relative to the question, not merely symbol count.

Question

Which expression most clearly says “any mixture of u ’s and v ’s”?

A regexp can state a negative local condition positively

Let A contain exactly the strings in $\{0, 1, 2\}^*$ with no substring 01. One description is

$$(1 \cup 2 \cup 0^+2)^*(0 \cup \epsilon)^*.$$

- Any 0 followed by more material must be followed by 0 or 2, never 1.
- Final 0's remain possible.

Guided analysis

Test 102, 1200, 2012, and 110.

Where we are

1. Regular expressions
2. Linear grammars
3. Regular relations

A linear grammar carries only one nonterminal at a time

Every rule has at most one nonterminal on its right-hand side:

$$A \rightarrow uBv \quad \text{or} \quad A \rightarrow w.$$

For example,

$$S \rightarrow aSb \mid \epsilon \quad \text{generates} \quad \{a^n b^n \mid n \geq 0\}.$$

Take-away

One continuing dependency can coordinate two terminal regions.

MOLR §4.4

Linear does not mean merely left-to-right

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$$

generates palindromes over $\{a, b\}$.

- Terminals accumulate on both sides of the single nonterminal.
- The dependency concerns paired choices, not linear distance alone.

Guided analysis

Derive *abbba*. What must change to derive only even-length palindromes?

Where we are

1. Regular expressions
2. Linear grammars
3. Regular relations

A regular relation describes aligned input–output pairs

$$R \subseteq \Sigma^* \times \Gamma^*$$

with atomic pairs and the same constructors used for regular expressions.

- The input and output alphabets need not be identical.
- Either side of an aligned step may be ϵ .
- The relation need not be a function unless each input has one output.

MOLR §4.4, regular relations

Rewrite notation can hide a relation

$$A \rightarrow B / C_D$$

describes correspondences between input strings containing A in the stated environment and output strings containing B .

- The rule abbreviates many paired strings.
- Finite-state locality depends on the form of C and D .
- A linguistic rule is empirical only after the alphabet and correspondence are interpreted.

Third hour: move among descriptions

- 1 Write a regexp for strings over $\{a, b\}$ with exactly one b .
- 2 Write a right-linear grammar for the same language.
- 3 Write a linear grammar for $\{a^n b^n \mid n \geq 0\}$.
- 4 Describe a relation that deletes every b .
- 5 Say which of these descriptions is functional and which are merely relational.

Week 7 synthesis

- 1 Regular expressions close atomic descriptions under choice, order, and repetition.
- 2 Equivalent descriptions need not expose the same generalization equally well.
- 3 Linear grammars preserve one continuing nonterminal dependency.
- 4 Regular relations describe paired strings, including insertion and deletion.
- 5 Description still precedes computation.

Question

Quiz 6 is the final pre-midterm check of descriptive vocabulary.